



CUSTOMER SEGMENTATION ANALYSIS USING RFM MODEL

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I. INTRODUCTION

Overview

The Marketing team is launching a **year-end promotion** targeting approximately **100,000 existing customers**. This report leverages **Customer 360 data** and applies **RFM analysis (Recency – Frequency – Monetary)** to segment customers by value and behavior, combined with the **BCG matrix** to evaluate potential. The goal is to identify priority customer groups and recommend **effective care strategies to maximize year-end revenue**

1. CUSTOMER 360

Customer 360 is a platform that helps businesses consolidate all customer data from various sources, including registration information, personal profiles, transaction history, and interaction behaviors. By gathering all data into a single system, businesses can track customer behavior comprehensively and make more accurate assessments of their value and engagement level.

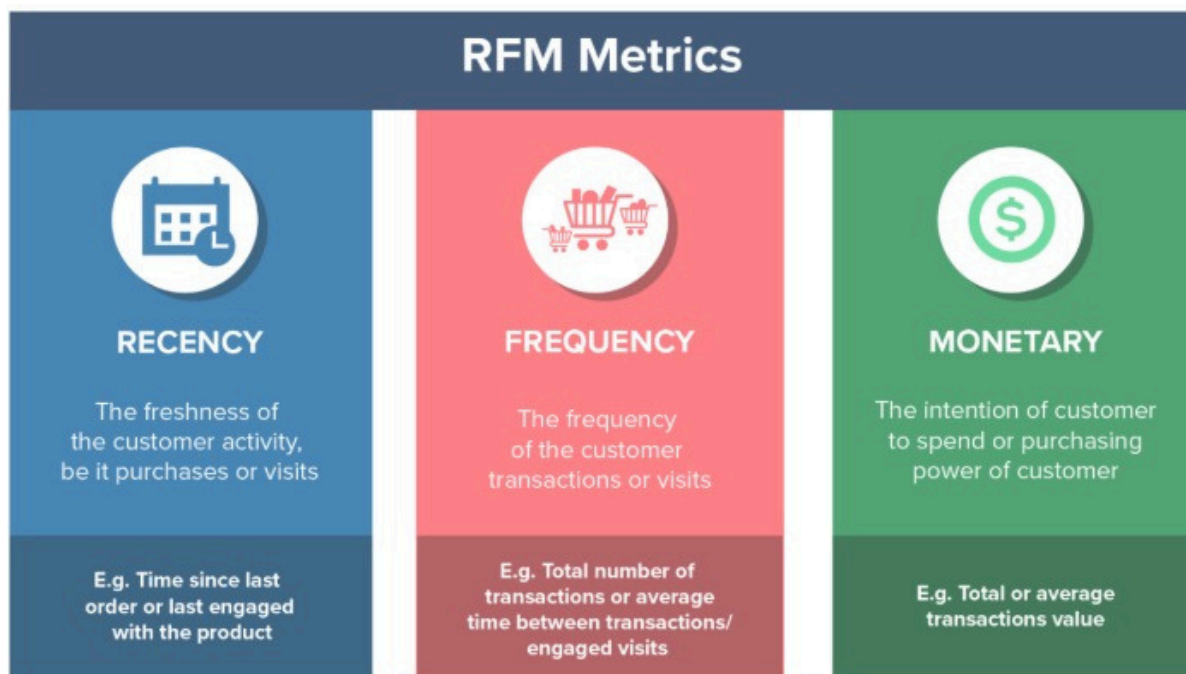


2. RFM MODEL

RFM is a customer analysis model based on three key factors:

- **Recency** – the time since the most recent transaction, reflecting current engagement.
- **Frequency** – the transaction frequency, indicating customer loyalty.
- **Monetary** – the spending value, reflecting the potential revenue contribution.

In Customer Analytics, RFM plays a crucial role as it allows businesses to quickly identify high-value customers, loyal customers, low-engagement customers, or those at risk of churn, thereby optimizing retention and reactivation strategies.

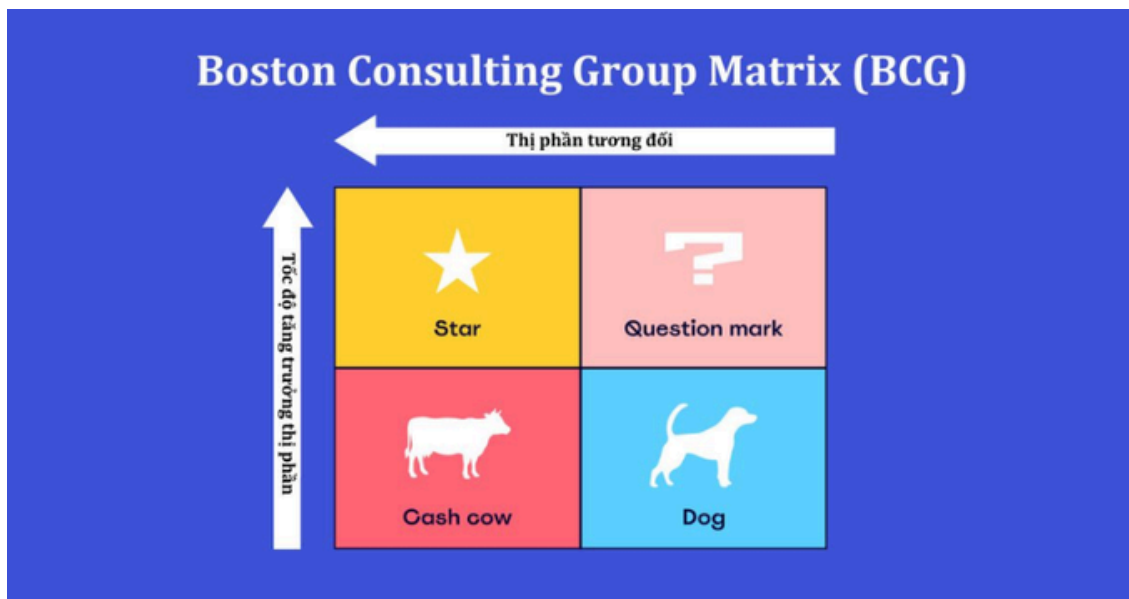


3. The Role of Combining RFM + BCG

Combining RFM segmentation results with the BCG matrix allows evaluation of customers from two perspectives:

- Current value (based on transaction behavior – RFM).
- Growth potential (based on BCG – contribution level and future development potential).

This approach not only identifies VIP, Loyal, or At-Risk groups but also indicates which groups should receive heavy investment, which ones should be maintained, and which can be cost-optimized. This supports Marketing in building effective customer care strategies and driving sustainable revenue growth.



II. DATA OVERVIEW AND STRUCTURE



Description of data sheets

Customer_Registered Table

The customer_registered table contains information about customers' service registrations, including key fields such as id (customer ID), contractID (contract ID), location (customer location), branchcode (branch code), status (contract status), and created_date (registration date). This table provides an important database for analyzing registration behavior, identifying the main serving branches, and tracking the operational status of customers.

Column Name	#	Type	Length	Scale	Precision	Not Null	Identity	Default
> 123 ID	1	float	8		53	[]	[]	
> A-Z Contract	2	varchar	50			[]	[]	
> 123 LocationID	3	real	4		24	[]	[]	
> 123 BranchCode	4	real	4		24	[]	[]	
> 123 Status	5	int	4		10	[]	[]	
> 2 created_date	6	datetime	8	3	23	[]	[]	
> 2 stopdate	7	datetime	8	3	23	[]	[]	

II. DATA OVERVIEW AND STRUCTURE



Description of data sheets

Customer_Transaction table

The customer_transaction table records customers' transaction history, including fields such as transactionID (transaction ID), customerID (customer ID), purchase_date (transaction date), and gmv (transaction value). This table serves as an important basis for analyzing shopping behavior, calculating RFM metrics (Recency, Frequency, Monetary), and assessing the value of each customer for care and marketing strategies.

Column Name	#	Type	Length	Scale	Precision	Not Null	Identity	Default
123 Transaction_ID	1	float	8		53	[]	[]	
123 CustomerID	2	float	8		53	[]	[]	
123 Purchase_Date	3	datetime	8	3	23	[]	[]	
123 GMV	4	float	8		53	[]	[]	



STEP 1: BUILDING THE CRFM TABLE

The CRFM table (Customer Recency-Frequency-Monetary) is created through a step-by-step process that transforms raw transactional data into meaningful RFM scores

Calculation of value: Recency, Frequency, Monetary

RFM is a customer analysis model based on three key factors:

- **Recency** – the time since the most recent transaction, reflecting current engagement.
- **Frequency** – the transaction frequency, indicating customer loyalty.
- **Monetary** – the spending value, reflecting the potential revenue contribution.

	123 CustomerID	123 created_date	123 Recency	123 Frequency	123 Monetary
1	0	2011-11-25 10:48:13.860	5	0.74	3,692.31
2	74,549	2015-06-01 15:12:03.590	62	0.14	17,241.38
3	74,819	2015-06-02 11:00:48.283	31	0.14	13,103.45
4	74,977	2015-06-02 15:43:38.037	92	0.14	15,862.07
5	75,284	2015-06-03 11:25:26.723	31	0.14	0
6	75,816	2015-06-04 15:25:55.250	92	0.14	18,620.69
7	76,217	2015-06-05 12:10:24.447	31	0.14	21,943.59
8	76,399	2015-06-05 17:32:49.130	92	0.14	9,655.17
9	76,551	2015-06-06 11:06:03.527	92	0.14	14,482.76
10	76,755	2015-06-06 16:06:08.973	62	0.14	21,943.59



STEP 2: RANKING AND QUARTILE DIVISION

RFM Scoring Method Using Quartiles:

The RFM scoring method is applied based on **Interquartile Range (IQR)**, in which each RFM metric is divided into four equally sized groups (Q1–Q4). This ensures that the customer base is evenly distributed, allowing comparison across segments with a balanced structure.

- **Recency:** Customers with more recent transactions are considered more active and valuable. Therefore, Recency is scored inversely — **lower Recency = higher score**. Customers in **Q1 (lowest Recency)** are assigned a score of **4**, while those in **Q4 (highest Recency)** receive a score of **1**.
- **Frequency & Monetary:** For both metrics, a **higher value indicates higher engagement and spending power**. Customers in **Q4 (highest Frequency/Monetary)** are assigned the maximum score of **4**, while customers in **Q1 (lowest values)** receive a score of **1**.

As a result, each customer will have three independent scores (R, F, M), each ranging from **1 to 4**, stored as categorical labels. This quartile-based scoring guarantees that **25% of customers fall into each score tier**, supporting a more **balanced segmentation distribution** and enabling fair comparisons across value segments.

123 CustomerID	123 Recency	123 Frequency	123 Monetary	AZ R	AZ F	AZ M
0	5	0.74	3,692.31	1	4	1
74,549	62	0.14	17,241.38	1	4	4
74,819	31	0.14	13,103.45	1	4	2
74,977	92	0.14	15,862.07	4	4	3
75,284	31	0.14	0	1	4	1
75,816	92	0.14	18,620.69	4	4	4
76,217	31	0.14	21,943.59	1	4	4
76,399	92	0.14	9,655.17	4	4	1
76,551	92	0.14	14,482.76	4	4	3
76,755	62	0.14	21,943.59	1	4	4
76,812	31	0.14	14,482.76	1	4	3
77,639	62	0.14	21,943.59	1	4	4
77,715	92	0.14	19,749.24	4	4	4
78,059	92	0.14	17,241.38	4	4	4
78,139	92	0.14	17,241.38	4	4	4
78,161	31	0.14	13,667.72	1	4	2
78,775	62	0.14	13,103.45	1	4	2
79,162	62	0.14	11,034.48	1	4	1
79,987	92	0.14	17,241.38	4	4	4



STEP3: COMBINE AND CLASSIFY

The table below summarizes how customers are classified based on their RFM score combinations. By grouping customers into Stars, Cash Cows, Question Marks, and Dogs, organizations can better understand customer value and define retention, growth, and reactivation strategies accordingly.

CUSTOMER SEGMENT	RFM SCORE	DESCRIPTION
STARS	431,432,434 ,443,444,44 1,442,433,3 44	Purchased recently - purchase frequently - spend a lot. (At least two scores = 4, and no score = 1) → VIP customers
CASH COWS	411,412,413 ,414,421,42 2,423,424,3 11,314	Purchased recently + lower purchasing frequency or lower spending (M = 1 or 2) + still spend relatively well → High-potential customers right after VIP
QUESTION MARKS	241,214,244 ,211,144,14 3	Haven't purchased for a long time (R = 1 or 2) + average frequency + average spending → "Old customers, not sure if they can be reactivated"
DOGS	131,132,133 ,134,141,14 2	Haven't purchased for a long time + purchase rarely + spend very little (no score = 4) → "Lost customers, not worth investing resources in"

III. CUSTOMER ANALYSIS

1. Overview

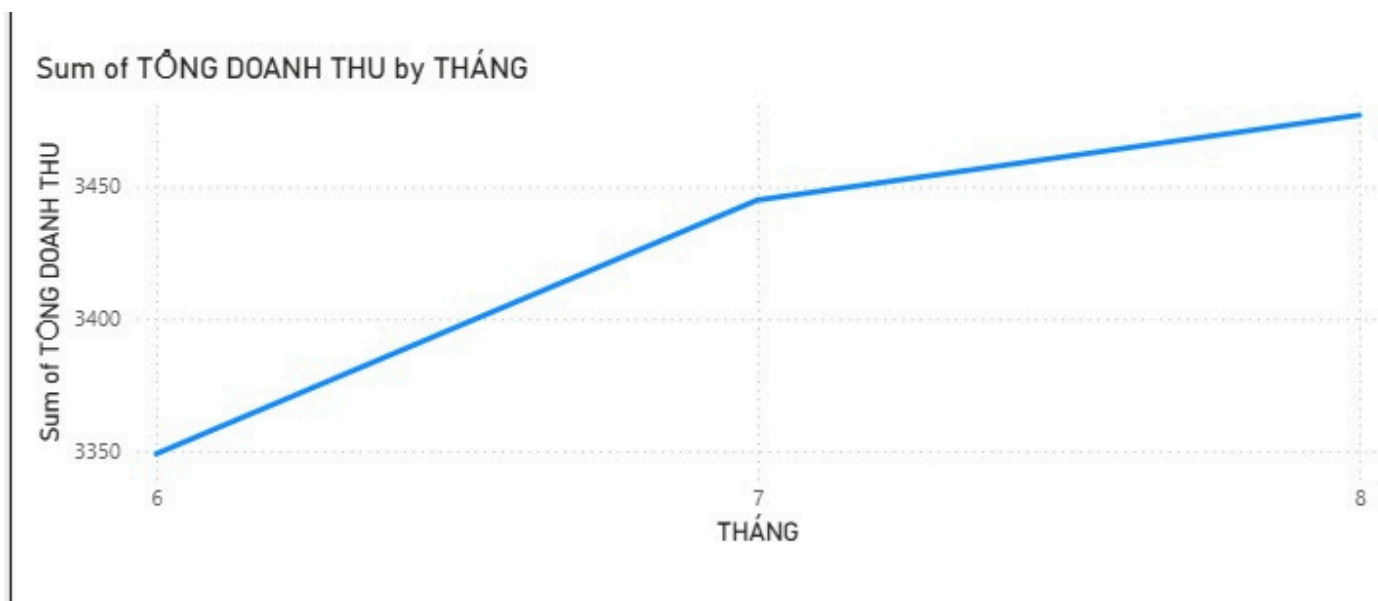
TOTAL CUSTOMER	TOTAL REVENUE (TOTAL GMV)	REVENUE PER CUSTOMER
130K	10.272M	79K

In the past three months (June to August), **revenue has shown a stable and positive growth trend**, increasing from 33.5 billion VND to 34.5 billion VND, **equivalent to a monthly growth rate of 3–4%.**

However, looking at the actual situation:

The business currently owns a large customer base of **130,000 people**, but the **total revenue generated is only 10,272 million VND**, and the **revenue per customer is just 79,000 VND** — a very low figure given the size of the customer base. This indicates that:

- **Most customers are new and tend to purchase only once** (possibly trial users or customers who buy during promotions).
- **The loyal customer rate is very low**, and the business has not fully leveraged the lifetime value of existing customers.



2. Customer segment analysis:

From the chart, it is evident that the current **RFM value distribution is severely imbalanced:**

- **The Question Marks and Dogs** segments dominate with extremely high proportions (**36.47% and 35.56% respectively, totaling over 72%**) – indicating that the company has a large number of customers who are either high-potential but **not yet effectively converted**, or have become “dormant” with minimal interaction and low revenue contribution. This shows that **the customer acquisition strategy** has been highly effective, yet the company is facing **significant difficulties in customer retention and stimulating repeat purchase frequency**.
- The second major concern is that **the Stars** segment accounts for **an extremely low share (only 4.44%)** – the near-complete absence of a core group of high-value, high-growth customers means the company **lacks a “successor force” to gradually replace the Cash Cows in the future**.

In summary, although short-term revenue growth remains supported by strong new customer acquisition, the current **RFM structure is sounding a critical red alert** regarding long-term sustainability. Without rapid and aggressive efforts to convert **Question Marks** and reduce the proportion of **Dogs**, the business risks **revenue stagnation or decline as the Cash Cows** segment continues to erode.

